

Answers

1.A	2.B	3.14	4.A	5.A	6.D	7.D	8.4
9.A	10.D	11.B	12.C	13.D	14.D	15.48	16.3
17.A	18.D	19.C	20.3	21.17	22.D	23.4	24.D
25.0	26.B	27.11	28.12	29.D	30.C	31.B	32.0
33.C	34.C	35.C	36.B	37.60	38.A	39.9	40.13
41.C	42.C	43.D	44.5	45.D	46.C	47.D	48.A
49.D	50.B	51.A	52.A	53.3	54.C	55.A	56.E
57.B	58.12	59.D	60.C	61.C	62.B	63.D	64.A
65.D	66.24	67.A	68.A	69.D	70.120	71.A	72.40
73.A	74.40	75.C					

Explanations

1. A

Let preeti obtain x and y in statistics and operation respectively. Now we know that $GPA = 3.2$.

So $(x+y+2+2) = 16$.

So $x+y = 12$. for this only one value is possible $x=y=6$.

So A grade.

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2. B

We are given that Party 3 received 70% of total pizzas, therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800 = 560$

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)						
Party 2 (120)						
Party 3 (560)						
Total (600)						

Total number of Thin Crust pizzas = $0.375 \times 800 = 300$. Therefore, total number of Deep Dish pizzas = $800 - 300 = 500$.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = $0.6 \times 120 = 72$. Consequently Party 1, must have received 42 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = $0.55 \times 120 = 66$. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$$\Rightarrow 300 - 72 - 66 = 162$$

Hence number of Deep Dish type of pizzas order by Party 3 = $560 - 162 = 398$

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)			72			48
Party 2 (120)			66			54
Party 3 (560)			162			398
Total (800)			300			500

Number of Normal Cheese pizzas require to be delivered to Party 2 = $0.3 \times 120 = 36$

It is given that 50% of these Normal Cheese pizzas were of Thin Crust variety, then We can say that remaining 50% were of Deep Dish variety. We can find out each of 4 types of pizzas require to be delivered to Party 2.

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)			72			48
Party 2 (120)	18	48	66	18	36	54
Party 3 (560)			162			398
Total (800)			300			500

Hence, the difference between the numbers of T-EC and D-EC pizzas to be delivered to Party 2 = $48 - 36 = 12$

Therefore, option B is the correct answer.

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3.14

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = $2 \times 20 + 16 + 14 = 70$

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = $70 - 10 = 60$.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = $60 - 2 \times 20 - 8 = 12$ which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited. Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60 - 8 - 20}{2} =$

16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = $2 \times 14 + 6 + 20 = 54$ {Assuming 100% marks in WE}. Since, Geeta was recruited at a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people (Jatin, Indu, Geeta, Ester) which were recruited. Hence, we can say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score is 52 or more.

Consequently, we can say that Geeta's composite score is either 52 or 53. Therefore we can say that Geeta scored either 18 $\{52 - (2 \times 14 + 6)\}$ or 19 $\{53 - (2 \times 14 + 6)\}$ marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$. Which is a possible case.

Case 2: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$ which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = $2 \times 8 + 19 + 16 = 51$ which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

It is given that all the candidates scored different marks in WE.

We can see that Ajay, Geeta and Ester have already scored 20, 19 and 18 marks in WE. Therefore, Harini can score a maximum of 17 marks in WE. If Harini's score in WE is 17, then Harini's composite score = $2 \times 5 + 17 + 20 = 47$ which is same as Falak's composite score. Hence, we can say that Harini can't score 17 marks in WE. Jatin and Danish have already scored 16 and 15 marks respectively.

Therefore, we can say that the maximum marks that Harini could have scored in WE = 14.

4. A

From given conditions we find that :

	London(l)	Paris(p)	Beijing(b)	NY(n)	Total
Round1(r1)	30	24	16	12	82
Round2(r2)	30	24(p r1) +3 (n r1) +4 (B r1) +1(n - representative)	12 (b r1) +9(n r1)	0	83
Round3(r3)	30+8 (Beijing r1 , r2)	24(p r1,r2) +4(b r1 ;p r2) +4(b r1,r2) +3 (n r1, p r2) +1 (n-representative) +1(b-representative)	0	0	75

We can see that - IOC member from New York must have voted for Paris in round 2.(IOC member - representative). Hence a is true. Also we can see that IOC member from Beijing voted for paris in round 3.Hence b is false. So answer is option A.

5. A

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be $4+5+5+a=14+a$, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = $1+2+5+5+b = 13+b$ (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = $3+5+5+c=13+c$ (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now $14+a=13+b=13+c$,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, $a=1, b=2$ and $c=2$. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1. So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1. Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. ($11 \times 2 + 1 = 23$)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12 ($4 + 4 + 4$). ($12 \times 2 + 1 = 25$)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

The highest score is 25.

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6.D

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be $4+5+5+a=14+a$, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = $1+2+5+5+b = 13+b$ (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = $3+5+5+c=13+c$ (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now $14+a=13+b=13+c$,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, $a=1, b=2$ and $c=2$. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1. So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1. Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. ($11 \times 2 + 1 = 23$)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12 ($4 + 4 + 4$). ($12 \times 2 + 1 = 25$)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Xyla was the highest scorer.

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7.D

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be $4+5+5+a=14+a$, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = $1+2+5+5+b = 13+b$ (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = $3+5+5+c=13+c$ (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now $14+a=13+b=13+c$,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, $a=1, b=2$ and $c=2$. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1. So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1. Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. ($11 \times 2 + 1 = 23$)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12 ($4 + 4 + 4$). ($12 \times 2 + 1 = 25$)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Tanzi scored 1 in round 3.

VIDEO SOLUTION

8.4

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Ezhil has Amudha, Dhinesh, Fani as his friends. Dhinesh has Bharatan as his friend. Hence a total of 4 (Amudha, Bharatan, Dhinesh, and Fani) are his friends or friend of a friend

 VIDEO SOLUTION

9. A

We are given that Party 3 received 70% of total pizzas, therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800 = 560$

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)						
Party 2 (120)						
Party 3 (560)						
Total (600)						

Total number of Thin Crust pizzas = $0.375 \times 800 = 300$. Therefore, total number of Deep Dish pizzas = $800 - 300 = 500$.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = $0.6 \times 120 = 72$. Consequently Party 1, must have received 42 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = $0.55 \times 120 = 66$. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$$\Rightarrow 300 - 72 - 66 = 162$$

Hence number of Deep Dish type of pizzas order by Party 3 = $560 - 162 = 398$

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)			72			48
Party 2 (120)			66			54
Party 3 (560)			162			398
Total (800)			300			500

Total number of Normal Cheese pizzas require to be delivered = $0.52 \times 800 = 416$

Number of Normal Cheese pizzas require to be delivered to Party 2 = $0.3 \times 120 = 36$

Number of Normal Cheese pizzas require to be delivered to Party 3 = $0.65 \times 560 = 364$

Therefore, total number of Normal Cheese pizzas require to be delivered to Party 1 = Total Normal Cheese pizzas to be delivered - Normal Cheese pizzas require to be delivered to Party 2 - Normal Cheese pizzas require to be delivered to Party 3

$$\Rightarrow 416 - 36 - 364 = 16$$

It is given that 25% of these 16 Normal Cheese pizzas were of Deep Dish type, hence the number of D- NC type pizza require to be delivered to Party 1 = $0.25 \times 16 = 4$

Consequently, the number of T- NC type pizza require to be delivered to Party 1 = $16 - 4 = 12$

We can find out each type of pizza that is required to be delivered to Party 1.

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)	12	60	72	4	44	48
Party 2 (120)			66			54
Party 3 (560)			162			398
Total (800)			300			500

Cost Price of a T-EC pizza = Rs. 500

Cost Price of a D-EC pizza = Rs. 550

Cost Price of a T-NC pizza = $\frac{3}{5} \times 550 = \text{Rs. } 330$

Cost Price of a D-NC pizza = $\frac{3}{5} \times 550 = \text{Rs. } 330$

Therefore the total bill amount for Party 1 = $12 \times 330 + 60 \times 500 + 4 \times 330 + 44 \times 550 = \text{Rs. } 59480$

Therefore, option A is the correct answer.

[VIDEO SOLUTION](#)

10. D

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = $2 \times 20 + 16 + 14 = 70$

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = $70 - 10 = 60$.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = $60 - 2 \times 20 - 8 = 12$ which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited. Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60 - 8 - 20}{2} = 16$

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = $2 \times 14 + 6 + 20 = 54$ {Assuming 100% marks in WE}. Since, Geeta was recruited at a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people (Jatin, Indu, Geeta, Ester) which were recruited. Hence, we can say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score is 52 or more.

Consequently, we can say that Geeta's composite score is either 52 or 53. Therefore we can say that Geeta scored either 18 { $52 - (2 \times 14 + 6)$ } or 19 { $53 - (2 \times 14 + 6)$ } marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$. Which is a possible case.

Case 2: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$ which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = $2 \times 8 + 19 + 16 = 51$ which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Let us check all the statement one by one.

Statement 1: Jatin's composite score was more than that of Danish. We can see that this statement is correct.

Statement 2: Indu scored less than Chetna in DI. We can see that Indu scored 16 marks in DI whereas Chetna scored 19 marks in DI. Hence, we can say that this statement is also correct.

Statement 3: Jatin scored more than Indu in GA. We can see that Jatin scored 14 marks in GA whereas Indu scored 20 marks in GA. Hence, we can say that this statement is incorrect.

Hence, we can say that option D is the correct answer.

[VIDEO SOLUTION](#)



11. **B**

Knowing the GPA we can find gawris grade in strategy as

$16 + x = 19$; $x = 3$. So her grade was C.

Now considering hari's grade $12 + x + y = 14$; so $x + y = 2$.

So he must have got D or F in strategy. Hence in Strategy, Gowri's grade point was higher than that obtained by hari.

[VIDEO SOLUTION](#)

12. **C**

Considering utkarsh's grades we get, $x + y = 6$. So possibilities of his missing grades are (4,2) or (6,0). Now according to given condition, his grades in marketing are same as faizal's in strategy and faizal got grade B in strategy. hence utkarsh got 4 in marketing and 2 in finance i.e. grade D. Hence option C.

[VIDEO SOLUTION](#)

13. **D**

The average income is calculated as show in the table given below.

Emp No.	Total Earning				Total Days				Average		
	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total	Complex	Medium	Simple
2001147	82.98	0	636.53	719.51	3	0	23	26	27.66	-	27.67522
2001148	51.53	0	461.73	513.26	3.3	1.67	16	21	15.61515	0	28.85813
2001149	171.10	0.61	79.10	250.81	5.5	4	8.5	18	31.10909	0.1525	9.305882
2001150	100.47	0.01	497.47	597.95	6	4.67	7.33	18	16.745	0.002141328	67.86767
2001151	594.43	159.64	0	754.06	9.67	13.33	0	23	61.47156	11.975994	-
2001156	83.70	0	6	89.70	8	0	1	9	10.4625	-	6
2001158	472.31	109.73	0	582.04	1.39	9.61	0	11	339.7914	11.41831426	-
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18	76.32827	60.91300746	318.9104
2001170	576.57	0	0	576.57	21	0	0	21	27.45571	-	-
2001171	286.47	6.10	0	292.57	8.38	4.25	0.38	13	34.18496	1.435294118	0
2001172	512.10	117.46	0	629.56	10	8.5	3.5	22	51.21	13.81882353	0
2001173	1303.88	0	0	1303.88	25.5	0	0.5	26	51.13255	-	0
2001174	1017.94	0	0	1017.9	26	0	0	26	39.15154	-	-
2001179	46.56	776.19	0	822.75	2	19	0	21	23.28	40.85210526	-
2001180	116.40	1262.79	0	1379.19	5	19	0	24	23.28	66.46263158	-

From the table given we can make out that number of employees whose average earning per day in complex operations is more than average earning per day in medium operations for 8 employees (2001148, 2001149, 2001150, 2001151, 2001158, 2001164, 2001171, 2001172)

[VIEW SOLUTION](#)

14. D

Let us consider that there are 'm' members totally, there are 4 cities so total number of voters in round 1 will be $m - 4$, as 4 members of the cities are not allowed to vote. In the same way we have in total $m - 3$ members who will vote in the 2nd round. For the 3rd round, we have $m - 2 - r$ members voting, r - number of voters who voted for NY and Beijing in Round 1 and 2. According to given conditions these voters can't vote again. Also, $m - 3 = 83$. So in total there were $m = 86$ members. And 83 people voted in 1st round and it is given that there are 75 votes in 3rd round. So we have $r = 9$ i.e. 9 voters who voted for NY in 2nd round and Beijing in 2nd round. Also it is given that NY voters in Round 1 voted either for Beijing or Paris and there are 9 voters who voted for NY in round 1, voted for Beijing in round 2. So $9 \times 100 / 12 = 75\%$. Hence option D.

[VIDEO SOLUTION](#)

15. 48

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. $I > F$

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $R > I$

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $I > P$

So basically there are two possible cases: $R > I > P > F$ or $R > I > F > P$

Case 1: Order of weights assigned = $R > I > P > F$

$R = 0.4, I = 0.3, P = 0.2, F = 0.1$

In this case overall score received by Best Ed = $0.1 \times 40 + 0.4 \times 30 + 0.2 \times 20 + 0.3 \times 20 = 26$

In this case overall score received by High Q = $0.1*30+0.4*20+0.2*20+0.3*40 = 27$

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = $R > I > F > P$

$R = 0.4, I = 0.3, P = 0.1, F = 0.2$

In this case overall score received by Best Ed = $0.2*40+0.4*30+0.1*20+0.3*20 = 28$

In this case overall score received by High Q = $0.2*30+0.4*20+0.1*20+0.3*40 = 28$

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table we can see that Education Aid scored the highest overall score = 48.

 VIDEO SOLUTION

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16.3

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. $I > F$

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $R > I$

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $I > P$

So basically there are two possible cases: $R > I > P > F$ or $R > I > F > P$

Case 1: Order of weights assigned = $R > I > P > F$

$R = 0.4, I = 0.3, P = 0.2, F = 0.1$

In this case overall score received by Best Ed = $0.1*40+0.4*30+0.2*20+0.3*20 = 26$

In this case overall score received by High Q = $0.1*30+0.4*20+0.2*20+0.3*40 = 27$

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = $R > I > F > P$

$R = 0.4, I = 0.3, P = 0.1, F = 0.2$

In this case overall score received by Best Ed = $0.2 \times 40 + 0.4 \times 30 + 0.1 \times 20 + 0.3 \times 20 = 28$

In this case overall score received by High Q = $0.2 \times 30 + 0.4 \times 20 + 0.1 \times 20 + 0.3 \times 40 = 28$

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table, we can see that three received the accreditation of AAA.

[VIDEO SOLUTION](#)

17. A

We need to find the person who has the highest average per-day earnings in medium operations.

So, we divide the total income in medium operations by the total number of day in these operations.

Employee no. 2001180 earned 1262.9 by working 19 days for medium operation which is highest. Hence option A.

[VIEW SOLUTION](#)

18. D

80% of attendance comes out to 20 days.

We can see from the table that there are 7 such employees who have earned more than Rs. 600 and having more than 80% of attendance.

Hence option D.

[VIEW SOLUTION](#)

19. C

Average income can be found by dividing the total income by total number of days.

The average income is calculated as show in the table given below.

Emp No.	Total Earning				Total Days				Average		
	Complex	Medium	Simple	Total	Complex	Medium	Simple	Total	Complex	Medium	Simple
2001147	82.98	0	636.53	719.51	3	0	23	26	27.66	-	27.67522
2001148	51.53	0	461.73	513.26	3.3	1.67	16	21	15.61515	0	28.85813
2001149	171.10	0.61	79.10	250.81	5.5	4	8.5	18	31.10909	0.1525	9.305882
2001150	100.47	0.01	497.47	597.95	6	4.67	7.33	18	16.745	0.002141328	67.86767
2001151	594.43	159.64	0	754.06	9.67	13.33	0	23	61.47156	11.975994	-
2001156	83.70	0	6	89.70	8	0	1	9	10.4625	-	6
2001158	472.31	109.73	0	582.04	1.39	9.61	0	11	339.7914	11.41831426	-
2001164	402.25	735.22	213.67	1351.14	5.27	12.07	0.67	18	76.32827	60.91300746	318.9104
2001170	576.57	0	0	576.57	21	0	0	21	27.45571	-	-
2001171	286.47	6.10	0	292.57	8.38	4.25	0.38	13	34.18496	1.435294118	0
2001172	512.10	117.46	0	629.56	10	8.5	3.5	22	51.21	13.81882353	0
2001173	1303.88	0	0	1303.88	25.5	0	0.5	26	51.13255	-	0
2001174	1017.94	0	0	1017.9	26	0	0	26	39.15154	-	-
2001179	46.56	776.19	0	822.75	2	19	0	21	23.28	40.85210526	-
2001180	116.40	1262.79	0	1379.19	5	19	0	24	23.28	66.46263158	-

Using this formula, we can find that there are 5 employees (Employee IDs: 1151, 1158, 1164, 1172, and 1173) who earned more than Rs 50 per day.

[VIEW SOLUTION](#)

20.3

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Ezhil has a total of 3 friends.



21.17

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2 . $\Rightarrow (1)$

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

$\Rightarrow$ The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 $\Rightarrow$ where a, b, c, ..., f are the unknown amounts raised.

Also $a + b + c + d + e + f = 21 - 2 = 19$.

We can observe that $19/6$ is slightly greater than 3 $\Rightarrow$ The average amount raised should be around 3.

If $a = 3$ and $f = 3 \Rightarrow b + c + d + e = 13$ (not possible) as the minimum case would be (4, 5, 6, 4) $\Rightarrow$ Not possible.

If $a = 3$ and $f = 2 \Rightarrow b + c + d + e = 14$ (not possible) as the minimum case would be (4, 5, 4, 3) $\Rightarrow$ Not possible.

$\Rightarrow a = 2$ and $f = 2 \Rightarrow b + c + d + e = 15$ the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If $a = 2$, b has to be equal to 3 to satisfy (1)

if $a = 3$, b has to be equal to 2 to satisfy (1)

$\Rightarrow$ The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between $\Rightarrow a + b = 7$ (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between $\Rightarrow a + b + c = 7$, the minimum case possible is 1, 2, 3, 2, 1 $\Rightarrow$ Satisfies.

Now, if there are 4 gaps $\Rightarrow a + b + c + d = 7 \Rightarrow$ The average value is $7/4$ which is less than 2 $\Rightarrow$ Not possible.

$\Rightarrow$ The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

$$\Rightarrow a + b + c = 8$$

When $a = 2$ and $c = 2 \Rightarrow b = 4 \Rightarrow 2, 4, 2 \Rightarrow$ Satisfies.

When $a = 2$ and $c = 3$, b should be 3 (Not satisfying (1))

When $a = 3$ and $c = 3$, b should be 2 (Not satisfying (1))

$\Rightarrow$ The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1, ..., 1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps $\Rightarrow a + b + c = 11$, the maximum case is 3, 5, 3 $\Rightarrow$ Satisfies.

Now, assume 4 gaps

$\Rightarrow a + b + c + d = 11$, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and $2 + 3 + 4 + 2 = 11$.

$\Rightarrow$ The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Maximum money raised in 2013 is $5 + 3 + 1 + 4 + 4 = 17$.

 VIDEO SOLUTION

22. D

	Round-1	Round-2	Round-3	Round-4	Round-5	Round-6
Tanzi	-	4	-	5	NP	NP
Umeza	-	-	-	1	2	NP
Wangdu	-	4	-	NP	NP	NP
Xyla	-	-	-	1	5	-
Yonita	-	-	3	5	NP	NP
Zeneca	-	-	-	5	5	NP

It is given that every bull's eye score in the first three rounds gave a player one additional chance to shoot in the bonus rounds, Rounds 4 to 6, which means Tanzi scored Bull's eye only once in the first 3 rounds because she participated only once in round 4 to 6. Similarly, Umeza scored Bull's eye exactly 2 times in the first 3 rounds. Wangdu did not score Bull's eye in the first three rounds and so on.

Now from 1, Tanzi, Umeza and Yonita had the same total score.

So, Total score of Tanzi will be $4+5+5+a=14+a$, (She scored Bull's eye(a score of 5) in exactly one round and a is the unknown score)

Total score of Umeza = $1+2+5+5+b = 13+b$ (She scored Bull's eye(a score of 5) in exactly 2 rounds and b is the unknown score)

Total score of Yonita = $3+5+5+c=13+c$ (She scored Bull's eye(a score of 5) in exactly one round and c is the unknown score)

Now $14+a=13+b=13+c$,

Also it is given that total scores for all players, except one, were in multiples of three, so these three will have to be a multiple of 3.

So, (a,b,c) can be either (1,2,2) or (4,5,5) in the same order. But the value (5,5) for b and c is not possible. (Umeza scored Bull's eye in exactly 2 rounds and Yonita in exactly 1 round)

Hence, $a=1, b=2$ and $c=2$. So each of Tanzi, Umeza and Yonita had total score of 15.

Tabulating the data, we have

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	1/5	4	1/5	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	-	-	-	5	5	NP	-

From 5, Tanzi and Zeneca had the same score in Round 1 but different scores in Round 3.

Zeneca score Bull's eye 2 times in round 1 to 3. If Tanzi scored 1 in round 1, then Zeneca also has to score 1 in round 1, which means both Tanzi and Zeneca scores in round 3 will be 5, which violates 5. Hence Tanzi scored 5 in round 1 and Zeneca also scored the same in round 1. So the new table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/5/2	5/5/2	5/5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2/5	2/5	3	5	NP	NP	15
Zeneca	5	5/(2/3/4)	5/(2/3/4)	5	5	NP	-

From 4, the number of players hitting bull's eye in Round 2 was double of that in Round 3.

So, in round 3 either 1 or 2 Bull's eye can be scored and in round 2, 2 or 4 Bull's eye can be scored.

Case 1: If only 1 Bull's eye is scored in the round 3, then in round 3 Umeza will score 2 and Zeneca will score 2/3/4 in round 3, which means both will score 5 in round 2. So minimum Bull's eye in round 2 will be 3. (Umeza, Zeneca and Xyla)

Hence this case is rejected.

Case 2: 2 Bull's eye were scored in round 3 and 4 Bull's eye were scored in round 2. So in round 2 Umeza, Yonita and Zeneca scored 5. This can be tabulated as:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	5/2	5	5/2	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	-

In round 3, 2 Bull's eye can only be scored by Xyla and Umeza.

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	1/2/3/4	4	1/2/3/4	NP	NP	NP	6-12
Xyla	5	5	5	1	5	1/2/3/4/5	22-26
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	(2/3/4)	5	5	NP	22-24

The highest scorer can be either Xyla or Zeneca. The lowest scorer will be Wangdu.

1. Consider Zeneca is the highest scorer.

From 3, the highest total score was one more than double of the lowest total score. So the only possible score for Zeneca is 23 and that for Wangdu is 11. ($11 \times 2 + 1 = 23$)

But this will violate condition 2, since both Zeneca and Wangdu do not have their scores as multiples of three in this case.

Hence, Xyla will be the highest scorer. The only possible total score for Xyla will be 25, and that for Wangdu is 12 ($4 + 4 + 4$). ($12 \times 2 + 1 = 25$)

Since Xyla already has non-multiple of 3 as total score. Zeneca will have 24 as the total score. The complete table is:

	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Total
Tanzi	5	4	1	5	NP	NP	15
Umeza	2	5	5	1	2	NP	15
Wangdu	4	4	4	NP	NP	NP	12
Xyla	5	5	5	1	5	4	25
Yonita	2	5	3	5	NP	NP	15
Zeneca	5	5	4	5	5	NP	24

Zeneca total score is 24.

VIDEO SOLUTION

23.4

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3, R4, R5 are $5 \times \text{means} = 17, 11, 19, 14$, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18$, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is 4 + 4)

=> We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => $2 + 4 + 4$ or $4 + 3 + 3$ (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as $3 + 4$ and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> Ratings give by R3 are 1, 4, 4, 5, 5 => Median = 4.

 VIDEO SOLUTION

24. D

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => $2x$ is the number of offline registrations => $3x$ is the total number of registrations.

According to the data given in the table => $3x$ should lie between the minimum and maximum total number of registrations. => $x = 40$ (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, ' x ' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The number of offline registrations in Feb is 50.

[VIDEO SOLUTION](#)

25.0

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3, R4, R5 are $5 \times \text{means} = 17, 11, 19, 14, \text{ and } 17$ respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18, \text{ and } 13$ respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is $4 + 4$)
=> We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ ($3 + 4$ or $4 + 3$) ----(1)

Consider R5, the missing elements should add up to 10 => $2 + 4 + 4$ or $4 + 3 + 3$ (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as $3 + 4$ and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> All ratings can be determined uniquely => 0.

 VIDEO SOLUTION

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26. B

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2 . => (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c, ..., f are the unknown amounts raised.

Also $a + b + c + d + e + f = 21 - 2 = 19$.

We can observe that $19/6$ is slightly greater than 3 => The average amount raised should be around 3.

If $a = 3$ and $f = 3$ => $b + c + d + e = 13$ (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If $a = 3$ and $f = 2$ => $b + c + d + e = 14$ (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> $a = 2$ and $f = 2$ => $b + c + d + e = 15$ the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If $a = 2$, b has to be equal to 3 to satisfy (1)

if $a = 3$, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => $a + b = 7$ (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => $a + b + c = 7$, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => $a + b + c + d = 7$ => The average value is $7/4$ which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> $a + b + c = 8$

When $a = 2$ and $c = 2$ => $b = 4$ => 2, 4, 2 => Satisfies.

When $a = 2$ and $c = 3$, b should be 3 (Not satisfying (1))

When $a = 3$ and $c = 3$, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => $a + b + c = 11$, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> $a + b + c + d = 11$, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and $2 + 3 + 4 + 2 = 11$.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

We see that only for C and D, we can conclude the amounts raised with certainty.

 VIDEO SOLUTION

27.11

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be $500000 - 275000 - 95000 - 85000 = 45000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 => the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be $100x$.

From 3, The winning candidate in constituency D must have got $5x + 37500$

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	100x
No. of votes polled by the winning candidate	2,75,000	48,750	140006	37500+5x
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

In constituency B, the mark for not losing the security deposit is $1/6(325000)$ or 54,167.

But winner himself/herself got $< 54167 \Rightarrow$ all the other candidates lost their security deposits.

11 is the correct answer,

[VIDEO SOLUTION](#)

28.12

The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.

We have the reference point of P1 at 66

Trying to fill in for C1:

The possible values are 75, 80, 85, 90, 95

Since 12 grams is of other nutrients, we can eliminate 90 and 95.

If we take 85, $85+12 = 97$, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, $75+12 = 87$ would leave 13 grams of protein, which, too, is eliminated.

Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.

Taking 85 carbs would give 2 grams of protein, which can be eliminated.

80 and 70 would also not work for the same reason.

Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100-65-14-8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100-70-14-8 = 8$ grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

The proteins in the foodgrains, when arranged in ascending order, are: 8, 8, 10, 12, 12, 14, 16

Giving the median as 12.

29. D

According to given condition 75% of voters who voted for Beijing in round 1, voted for Beijing in round 2. There are $21-9 = 12$ voters who voted for Beijing in round 2 also Beijing in round 1. Hence total voters for Beijing in round 1 is 16. Son in round 1 total 82 voters, out of which Beijing -16 voters, London - 30, New York - 12. Hence total voters for Paris $82-16-30-12 = 24$. Hence option D.

30. C

We are given that Party 3 received 70% of total pizzas, therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800 = 560$

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)						
Party 2 (120)						
Party 3 (560)						
Total (600)						

Total number of Thin Crust pizzas = $0.375 \times 800 = 300$. Therefore, total number of Deep Dish pizzas = $800 - 300 = 500$.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = $0.6 \times 120 = 72$. Consequently Party 1, must have received 48 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = $0.55 \times 120 = 66$. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$$\Rightarrow 300 - 72 - 66 = 162$$

Hence number of Deep Dish type of pizzas order by Party 3 = $560 - 162 = 398$

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)			72			48
Party 2 (120)			66			54
Party 3 (560)			162			398
Total (800)			300			500

Total number of Normal Cheese pizzas require to be delivered = $0.52 \times 800 = 416$

Number of Normal Cheese pizzas require to be delivered to Party 2 = $0.3 \times 120 = 36$

Number of Normal Cheese pizzas require to be delivered to Party 3 = $0.65 \times 560 = 364$

Therefore, total number of Normal Cheese pizzas require to be delivered to Party 1 = Total Normal Cheese pizzas to be delivered - Normal Cheese pizzas require to be delivered to Party 2 - Normal Cheese pizzas require to be delivered to Party 3

$$\Rightarrow 416 - 36 - 364 = 16$$

Hence, option C is the correct answer.



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31. B

We are given that Party 3 received 70% of total pizzas, therefore, number of pizzas received by Party 3 = $\frac{70}{100} \times 800 = 560$

Remaining 240 pizzas are equally divided among party 1 and party 2 hence we can say that each of Party 1 and Party 2 received 120 pizzas.

We know that all of the pizza can be classified into a total of 4 types. Hence, on drawing a table which can accommodate all of the cases:

	Thin Crust (T)			Deep Dish (D)		
	T - NC	T - EC	Total	D - NC	D - EC	Total
Party 1 (120)						
Party 2 (120)						
Party 3 (560)						
Total (800)						

Total number of Thin Crust pizzas = $0.375 \times 800 = 300$. Therefore, total number of Deep Dish pizzas = $800 - 300 = 500$.

Out of 120 pizzas that Party 1 received, 60% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 1 = $0.6 \times 120 = 72$. Consequently Party 1, must have received 48 Deep Dish type pizzas.

Out of 120 pizzas that Party 2 received, 55% were of Thin Crust type hence, total number of Thin Crust pizza received by Party 2 = $0.55 \times 120 = 66$. Consequently Party 1, must have received 54 Deep Dish type pizzas.

Therefore, total number of Thin Crust pizzas ordered by Party 3 = Total Thin Crust pizzas ordered - Thin Crust pizzas ordered by Party 1 - Thin Crust pizzas ordered by Party 2

$$\Rightarrow 300 - 72 - 66 = 162$$

Hence, option B is the correct answer.

32. 0

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

$\Rightarrow$ The sum of ratings given by R1, R2, R3, R4, R5 are $5 \times \text{means} = 17, 11, 19, 14, \text{ and } 17$ respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18, \text{ and } 13$ respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is 4 + 4)

=> We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

R2 gave ratings of 1, 1, 2, 2, 5 => He gave 4 to 0 workers => 0 is the answer.

 VIDEO SOLUTION

33. C

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3, R4, R5 are $5 \times \text{means} = 17, 11, 19, 14$, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18$, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is 4 + 4)

=> We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1		4			11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => 2 + 4 + 4 or 4 + 3 + 3 (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

=> R2 median rating is 2 => given to 2 workers

=> R5 median rating is 4 => given to 2 workers

=> R4 median rating is 3 => given to only 1 worker.

=> R3 median rating is 4 => given to 2 workers.

 VIDEO SOLUTION

34. C

Since Aditya didn't get a call from any of the colleges, so for each college, he either didn't clear one of the sectional cut-offs or he didn't clear the aggregate cutoff or both. If he didn't clear one of the sectional cut-offs, then for that section he scored less marks than the least cut-off among the given cut-offs of all the colleges.

For example, for section A, it is given that the cut-offs for colleges 1, 4 and 5 are 42, 43 and 45 respectively. The least cut-off among them is 42. So, in order to not clear the sectional cut-off of section A for colleges 1, 4 and 5, he should have scored less than 42.

Similarly for section B, in order to not clear the sectional cut-off for colleges 1, 2 and 6, he should have scored less than 41.

Similarly for section C, in order to not clear the sectional cut-off for colleges 1, 2, 3 and 5, he should have scored less than 42.

Similarly for section D, in order to not clear the sectional cut-off for colleges 4 and 6, he should have scored less than 44.

We can see that if we consider section C and D then we will cover all six colleges. So the maximum score that one can score in section C and D despite that he misses the sectional cut off are 41 and 43 respectively. Also in order to maximize the total we can assume that the person scored full marks in the both section A and B.

Maximum marks obtained by Aditya such that he doesn't get any calls = $41 + 43 + 50 + 50 = 184$

Hence, option C is the correct answer.

 VIDEO SOLUTION

35. C

We have to minimum marks obtained by him in a section which is possible only when Charlie must have scored full marks in remaining 3 sections. Along with that we have to ensure that Charlie gets enough marks to clear aggregate cut off as well.

For college 1, the minimum marks that Charlie needs to get a call = 176. To get the minimum marks in any section he has to score full marks in remaining 3 sections. Since he has to clear the cut off in sections A, B and C we can say that he can score a minimum marks in section D.

Minimum marks in section D despite which he will get call from college 1 = $176 - 3 \times 50 = 26$.

Similarly, minimum marks in one section out of A and D despite which he will get call from college 2 = $175 - 3 \times 50 = 25$.

Similarly, minimum marks in one section out of A, B and D despite which he will get call from college 3 = $171 - 3 \times 50 = 21$.

Similarly, minimum marks in one section out of B and C despite which he will get call from college 4 = $178 - 3 \times 50 = 28$.

Similarly, minimum marks in one section out of B and D despite which he will get call from college 5 = $180 - 3 \times 50 = 30$.

Similarly, minimum marks in one section out of A and C despite which he will get call from college 6 = $176 - 3 \times 50 = 26$.

We can see that Charlie need to score at least 25 marks in one section out of A and D to get call from exactly 2 colleges. Therefore, option C is correct answer.

 VIDEO SOLUTION

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36. B

We can see that Bhama got calls from all the college hence it is essential for her to clear all 4 sectionals along with overall cutoff. That essentially means that she has to score marks which are greater or equal to the greatest cut-off for that section across the six colleges. from the table we can see that to clear section A the cut offs for colleges 1, 4 and 5 are 42, 43 and 45 respectively. Hence, in order to clear the sectional cut-off of section A for all the colleges, she should have scored at least 45 marks.

Similarly, the minimum marks required sectional cut-off for section B for all colleges = $\text{Max}\{42, 42, 41\} = 45$.

Similarly, the minimum marks required sectional cut-off for section C for all colleges = $\text{Max}\{42, 45, 46, 43\} = 46$.

Similarly, the minimum marks required sectional cut-off for section D for all colleges = $\text{Max}\{45, 44\} = 45$.

So total minimum aggregate marks required is $45+45+46+45 = 181$.

Note we have to check for aggregate cut-off as well for all 6 colleges. We can see that 181 is also more than aggregate cut-off marks of all six colleges. Therefore option B is the correct answer.

 VIDEO SOLUTION

37.60

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test $\Rightarrow$ Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$$\Rightarrow 60x = 24$$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*0.4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $\{66-70*(0.6)\}/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the score obtained by Rini in the project is 60

[VIDEO SOLUTION](#)

38. A

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = $2*20+16+14 = 70$

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = $70 - 10 = 60$.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = $60 - 2*20 - 8 = 12$ which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited. Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60 - 8 - 20}{2} = 16$

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = $2*14 + 6 + 20 = 54$ {Assuming 100% marks in WE}. Since, Geeta was recruited at a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people (Jatin, Indu, Geeta, Ester) which were recruited. Hence, we can say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score is 52 or more.

Consequently, we can say that Geeta's composite score is either 52 or 53. Therefore we can say that Geeta scored either 18 $\{52 - (2 \times 14 + 6)\}$ or 19 $\{53 - (2 \times 14 + 6)\}$ marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$. Which is a possible case.

Case 1: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$ which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = $2 \times 8 + 19 + 16 = 51$ which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Let us check all the statements one by one.

Option A: Bala scored same as Jatin in DI. We can say that Bala scored 20 marks in DI. In that Bala's composite score = $2 \times 20 + 9 + 11 = 60$ which is same as Indu's composite score. Therefore, we can say that this is a false statement. Hence, option A is the correct answer.

 VIDEO SOLUTION

39.9

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95,000 votes, hence the second runner up will get 85,000 votes.

Now the remaining votes will be $500,000 - 275,000 - 95,000 - 85,000 = 45,000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 $\Rightarrow$ the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be $100x$.

From 3, The winning candidate in constituency D must have got $15x+37500$

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	$100x$
No. of votes polled by the winning candidate	2,75,000	48,750	140006	$37500+5x$
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

Total votes in A = 500000 $\Rightarrow$ the candidates who got ≤ 83333 must have lost their security deposits $\Rightarrow$ candidates till 2nd runner up didn't lose their deposit $\Rightarrow$ all the candidates who received $500000-275000-95000-85000=45000$ lost their deposits.

The percentage of votes polled in total by all the candidates who lost their security deposits while contesting for constituency A = $45000 \times 100 / 500000 = 9\%$

[VIDEO SOLUTION](#)

40.13

It is given that Indu and Jatin both scored 100% in exactly one section. We can say that Jatin scored 100% marks in DI. Therefore, Jatin's composite score = $2 \times 20 + 16 + 14 = 70$

It is given that Jatin's composite score was 10 more than Indu's. Therefore, we can say that Indu's composite score = $70 - 10 = 60$.

Indu also scored 100% in exactly one section.

Case 1: Indu scored 100% marks in DI.

If Indu scored 100% marks in DI, then Indu's score in GA = $60 - 2 \times 20 - 8 = 12$ which is less than 70% of maximum possible marks. Indu already has less than 70% in WE, therefore we Indu can't be recruited. Hence, we can reject this case.

Consequently, we can say that Indu scored 100% marks in WE. Therefore, Indu's score in DI = $\frac{60 - 8 - 20}{2} =$

16

It is also given that Danish, Harini, and Indu had scored the same marks the in GA.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14		6	
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

We are given that, among the four recruited, Geeta had the lowest composite score.

Maximum composite score that Geeta can get = $2 \times 14 + 6 + 20 = 54$ {Assuming 100% marks in WE}. Since, Geeta was recruited at a composite score of 54 or less we can say that Ester was definitely recruited.

It is given that no two candidates had the same composite score. We can see that Chetna's composite score is 54. Hence, Geeta can't have a composite score of 54. Therefore, we can say that Geeta's composite score is 53 or less.

We already know the four people (Jatin, Indu, Geeta, Ester) which were recruited. Hence, we can say that Danish was rejected at a composite score of 51. Hence, we can say that Geeta's composite score is 52 or more.

Consequently, we can say that Geeta's composite score is either 52 or 53. Therefore we can say that Geeta scored either 18 $\{52 - (2 \times 14 + 6)\}$ or 19 $\{53 - (2 \times 14 + 6)\}$ marks in WE.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8		16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	18/19	6	52/53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

Ajay was the unique highest scorer in WE.

Case 1: Geeta scored 19 marks in WE.

We can say that if Geeta scored 19 marks in WE, then Ajay scored 20 marks in DI. In that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$. Which is a possible case.

Case 2: Geeta scored 18 marks in WE.

We can say that if Geeta scored 18 marks in WE, then Ajay can score either 19 or 20 marks in DI.

If Ajay scored 20 marks in DI then in that case Ajay's composite score = $2 \times 8 + 20 + 16 = 52$ which will be same as Geeta's composite score. Hence, we can say that in this case Ajay can't score 20 marks.

If Ajay scored 19 marks in DI then in that case Ajay's composite score = $2 \times 8 + 19 + 16 = 51$ which will be same as Danish's composite score. Hence, we can say that in this case Ajay can't score 19 marks.

Therefore, we can say that case 2 is not possible at all.

Candidate	Marks out of 20			Composite score
	DI	WE	GA	
Ajay	8	20	16	52
Bala		9	11	
Chetna	19	4	12	54
Danish	8	15	20	51
Ester	12	18	16	58
Falak	15	7	10	47
Geeta	14	19	6	53
Harini	5		20	
Indu	16	8	20	60
Jatin	20	16	14	70

It is given that all the candidates except Ajay and Danish had different marks in DI and Bala's composite score was less than Chetna's composite score.

Let us assume that Bala scored 'x' marks in DI.

$$\Rightarrow 2x + 9 + 11 < 54$$

$$\Rightarrow x < 17$$

We can see that Bala's score will be less than 17. Bala's maximum score in DI will be the largest possible number less than 17 which is not same as any other candidate's score in DI. From the table we can see that 16, 15 and 14 are already taken by Indu, Falak and Geeta respectively.

Therefore, we can say that Bala can score a maximum of 13 marks in DI.

[VIDEO SOLUTION](#)

Free CAT Study Material

41. C

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2 . $\Rightarrow (1)$

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

$\Rightarrow$ The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 $\Rightarrow$ where a, b, c, ..., f are the unknown amounts raised.

Also $a + b + c + d + e + f = 21 - 2 = 19$.

We can observe that $19/6$ is slightly greater than 3 $\Rightarrow$ The average amount raised should be around 3.

If $a = 3$ and $f = 3 \Rightarrow b + c + d + e = 13$ (not possible) as the minimum case would be (4, 5, 6, 4) $\Rightarrow$ Not possible.

If $a = 3$ and $f = 2 \Rightarrow b + c + d + e = 14$ (not possible) as the minimum case would be (4, 5, 4, 3) $\Rightarrow$ Not possible.

$\Rightarrow a = 2$ and $f = 2 \Rightarrow b + c + d + e = 15$ the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If $a = 2$, b has to be equal to 3 to satisfy (1)

if $a = 3$, b has to be equal to 2 to satisfy (1)

$\Rightarrow$ The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between $\Rightarrow a + b = 7$ (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between $\Rightarrow a + b + c = 7$, the minimum case possible is 1, 2, 3, 2, 1 $\Rightarrow$ Satisfies.

Now, if there are 4 gaps $\Rightarrow a + b + c + d = 7 \Rightarrow$ The average value is $7/4$ which is less than 2 $\Rightarrow$ Not possible.

$\Rightarrow$ The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

$$\Rightarrow a + b + c = 8$$

When a = 2 and c = 2 $\Rightarrow$ b = 4 $\Rightarrow$ 2, 4, 2 $\Rightarrow$ Satisfies.

When a = 2 and c = 3, b should be 3 (Not satisfying (1))

When a = 3 and c = 3, b should be 2 (Not satisfying (1))

$\Rightarrow$ The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1, ..., 1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps $\Rightarrow a + b + c = 11$, the maximum case is 3, 5, 3 $\Rightarrow$ Satisfies.

Now, assume 4 gaps

$\Rightarrow a + b + c + d = 11$, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and $2 + 3 + 4 + 2 = 11$.

$\Rightarrow$ The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

C	2013	2014	2015	2016	2017
	1	2	3	2	1

D	2011	2012	2013	2014	2015
	1	2	4	2	1

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

Money raised in 2015 is $2 + 1 + 3 + 1 + 0/1 = 7$ or 8 .

 VIDEO SOLUTION

42. C

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2 . $\Rightarrow$ (1)

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

=> The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 => where a, b, c, ..., f are the unknown amounts raised.

Also $a + b + c + d + e + f = 21 - 2 = 19$.

We can observe that $19/6$ is slightly greater than 3 => The average amount raised should be around 3.

If $a = 3$ and $f = 3$ => $b + c + d + e = 13$ (not possible) as the minimum case would be (4, 5, 6, 4) => Not possible.

If $a = 3$ and $f = 2$ => $b + c + d + e = 14$ (not possible) as the minimum case would be (4, 5, 4, 3) => Not possible.

=> $a = 2$ and $f = 2$ => $b + c + d + e = 15$ the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If $a = 2$, b has to be equal to 3 to satisfy (1)

if $a = 3$, b has to be equal to 2 to satisfy (1)

=> The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between => $a + b = 7$ (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between => $a + b + c = 7$, the minimum case possible is 1, 2, 3, 2, 1 => Satisfies.

Now, if there are 4 gaps => $a + b + c + d = 7$ => The average value is $7/4$ which is less than 2 => Not possible.

=> The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

=> $a + b + c = 8$

When $a = 2$ and $c = 2$ => $b = 4$ => 2, 4, 2 => Satisfies.

When $a = 2$ and $c = 3$, b should be 3 (Not satisfying (1))

When $a = 3$ and $c = 3$, b should be 2 (Not satisfying (1))

=> The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1,.....,1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps => $a + b + c = 11$, the maximum case is 3, 5, 3 => Satisfies.

Now, assume 4 gaps

=> $a + b + c + d = 11$, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and $2 + 3 + 4 + 2 = 11$.

=> The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1
B	2012	2013	2014	2015				
	1	2	3	1				
	1	3	2	1				
C	2013	2014	2015	2016	2017			
	1	2	3	2	1			
D	2011	2012	2013	2014	2015			
	1	2	4	2	1			
E	2010	2011	2012	2013	2014	2015		
	1	3	5	3	1	-		
	1	2	3	4	2	1		
	1	2	4	3	2	1		

Given that E raised 3 in 2013 => in 2012 he could have raised a minimum of 4 crores.

=> Minimum amount is $4 + 1 + 0 + 2 + 4 = 11$.

 VIDEO SOLUTION

43. D

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => $2x$ is the number of offline registrations => $3x$ is the total number of registrations.

According to the data given in the table => $3x$ should lie between the minimum and maximum total number of registrations. => $x = 40$ (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, ' x ' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

1) In May, there are 30 offline registrations (smallest) => True

2) In Mar, we have smallest number of total registrations => False.

[VIDEO SOLUTION](#)

44.5

The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.

We have the reference point of P1 at 66

Trying to fill in for C1:

The possible values are 75, 80, 85, 90, 95

Since 12 grams is of other nutrients, we can eliminate 90 and 95.

If we take 85, $85 + 12 = 97$, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, $75 + 12 = 87$ would leave 13 grams of protein, which, too, is eliminated.

Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.

Taking 85 carbs would give 2 grams of protein, which can be eliminated.

80 and 70 would also not work for the same reason.

Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that all other foodgrains except M3 had more carbohydrates than M1. Therefore, 5 is the correct answer.

[VIDEO SOLUTION](#)

45. D

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes. Now the remaining votes will be $500000 - 275000 - 95000 - 85000 = 45000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 $\Rightarrow$ the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

Let the number of votes polled in D be $100x$.

From 3, The winning candidate in constituency D must have got $37500 + 5x$.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	$100x$
No. of votes polled by the winning candidate	2,75,000	48,750	140006	$37500 + 5x$
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

Number of votes polled to winning candidate must be 140006.

[VIDEO SOLUTION](#)

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46. C

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes. Now the remaining votes will be $500000 - 275000 - 95000 - 85000 = 45000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 $\Rightarrow$ the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

From 3, The winning candidate in constituency D must have got $1.05 \times 37500 = 39375$.

Let the number of votes polled in D be $100x$.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	$100x$
No. of votes polled by the winning candidate	2,75,000	48,750	140006	39,375
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

As calculated in the previous question, candidate D got 175000 votes. The winner got 5% more votes than first runner up, hence the winner got 0.05×175000 i.e 8750 more votes than first runner up . Thus 8750 is the winning margin for constituency D. Moreover margin in constituency is atleast 10000 . Hence in the increasing order C will always come after D which is not happening in the third option. Hence that is the correct answer.

 VIDEO SOLUTION

47. D

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. $I > F$

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $R > I$

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $I > P$

So basically there are two possible cases: $R > I > P > F$ or $R > I > F > P$

Case 1: Order of weights assigned = $R > I > P > F$

$R = 0.4, I = 0.3, P = 0.2, F = 0.1$

In this case overall score received by Best Ed = $0.1 \times 40 + 0.4 \times 30 + 0.2 \times 20 + 0.3 \times 20 = 26$

In this case overall score received by High Q = $0.1 \times 30 + 0.4 \times 20 + 0.2 \times 20 + 0.3 \times 40 = 27$

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = $R > I > F > P$

$R = 0.4, I = 0.3, P = 0.1, F = 0.2$

In this case overall score received by Best Ed = $0.2 \times 40 + 0.4 \times 30 + 0.1 \times 20 + 0.3 \times 20 = 28$

In this case overall score received by High Q = $0.2 \times 30 + 0.4 \times 20 + 0.1 \times 20 + 0.3 \times 40 = 28$

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

We can see that weight of the faculty quality parameter = 0.1. Hence, option D is the correct answer.

VIDEO SOLUTION

48. A

It is given that: High Q > Best Ed > Cosmopolitan and Education Aid > A-one

We can say that High Q > Cosmopolitan

We can see that both High Q and Cosmopolitan got same points in reputation (R) and placement quality (P). High Q received more points in infrastructure (I) than Cosmopolitan whereas Cosmopolitan received more points in faculty Quality (F) than High Q.

Hence, we can say that Infrastructure's weight should be greater than Faculty quality. i.e. $I > F$

Similarly, We can see that both Best Ed and Cosmopolitan got same points in faculty Quality (F) and placement quality (P). Best Ed received more points in reputation (R) than Cosmopolitan whereas Cosmopolitan received more points in infrastructure (I) than Best Ed.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $R > I$

Similarly, We can see that both Education Aid and A-one got same points in faculty Quality (F) and reputation (R). Education Aid received more points in infrastructure (I) than A-one whereas A-one received more points in placement quality (P) than Education Aid.

Hence, we can say that reputation's weight should be greater than infrastructure. i.e. $I > P$

So basically there are two possible cases: $R > I > P > F$ or $R > I > F > P$

Case 1: Order of weights assigned = $R > I > P > F$

$R = 0.4, I = 0.3, P = 0.2, F = 0.1$

In this case overall score received by Best Ed = $0.1 \times 40 + 0.4 \times 30 + 0.2 \times 20 + 0.3 \times 20 = 26$

In this case overall score received by High Q = $0.1 \times 30 + 0.4 \times 20 + 0.2 \times 20 + 0.3 \times 40 = 27$

We can see that High Q's overall score is higher than Best Ed. Hence, this is a possible case.

Case 2: Order of weights assigned = $R > I > F > P$

$R = 0.4, I = 0.3, P = 0.1, F = 0.2$

In this case overall score received by Best Ed = $0.2 \times 40 + 0.4 \times 30 + 0.1 \times 20 + 0.3 \times 20 = 28$

In this case overall score received by High Q = $0.2 \times 30 + 0.4 \times 20 + 0.1 \times 20 + 0.3 \times 40 = 28$

We can see that High Q's overall score is not greater than the overall score received Best Ed. Hence, this case is not possible.

Now that we know the weight of each parameter, we can calculate the overall score and accreditation received by each college.

	F (0.1)	R (0.4)	P (0.2)	I (0.3)	Overall score	Accreditation
A-one	50	50	50	40	47	AAA
Best Ed	40	30	20	20	26	BBA
Cosmopolitan	40	20	20	30	25	BBA
Dominance	20	20	40	30	27	BBA
Education Aid	50	50	40	50	48	AAA
Fancy	50	50	40	40	45	AAA
Global	30	0	20	20	13	Junk
High Q	30	20	20	40	27	BBA

From the table we can see that none of the mentioned college received an overall scores between 31 and 40, both inclusive. Hence, option A is the correct answer.

VIDEO SOLUTION

49. D

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharathan are known the strangers are found for Bharathan.

Since Fani, Gowtham are strangers to Amudha, Bharathan. Amudha, Bharathan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

In the given options Bharatan and Ezhil are Acauaintances, Fani and Gowtham are acquaintances.

50. B

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C. Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharatan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Fani is an acquaintance of Chandran

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51. A

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be ± 1 or ± 2 . $\Rightarrow (1)$

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

$\Rightarrow$ The raising pattern looks like follows:

1, a, b, c, d, e, f, 1 $\Rightarrow$ where a, b, c, ..., f are the unknown amounts raised.

Also $a + b + c + d + e + f = 21 - 2 = 19$.

We can observe that $19/6$ is slightly greater than 3 $\Rightarrow$ The average amount raised should be around 3.

If $a = 3$ and $f = 3 \Rightarrow b + c + d + e = 13$ (not possible) as the minimum case would be (4, 5, 6, 4) $\Rightarrow$ Not possible.

If $a = 3$ and $f = 2 \Rightarrow b + c + d + e = 14$ (not possible) as the minimum case would be (4, 5, 4, 3) $\Rightarrow$ Not possible.

$\Rightarrow a = 2$ and $f = 2 \Rightarrow b + c + d + e = 15$ the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1

Consider B:

The patterns looks as follows:

1, a, b, 1

If $a = 2$, b has to be equal to 3 to satisfy (1)

if $a = 3$, b has to be equal to 2 to satisfy (1)

$\Rightarrow$ The possible cases for B are:

B	2012	2013	2014	2015
	1	2	3	1
	1	3	2	1

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between $\Rightarrow a + b = 7$ (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between $\Rightarrow a + b + c = 7$, the minimum case possible is 1, 2, 3, 2, 1 $\Rightarrow$ Satisfies.

Now, if there are 4 gaps $\Rightarrow a + b + c + d = 7 \Rightarrow$ The average value is $7/4$ which is less than 2 $\Rightarrow$ Not possible.

$\Rightarrow$ The possible cases for C are:

C	2013	2014	2015	2016	2017
	1	2	3	2	1

Consider D:

The pattern looks as follows:

1, a, b, c, 1

$$\Rightarrow a + b + c = 8$$

When $a = 2$ and $c = 2 \Rightarrow b = 4 \Rightarrow 2, 4, 2 \Rightarrow$ Satisfies.

When $a = 2$ and $c = 3$, b should be 3 (Not satisfying (1))

When $a = 3$ and $c = 3$, b should be 2 (Not satisfying (1))

$\Rightarrow$ The possible cases for D are:

D	2011	2012	2013	2014	2015
	1	2	4	2	1

Consider E:

The pattern looks as follows:

1, ..., 1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps $\Rightarrow a + b + c = 11$, the maximum case is 3, 5, 3 $\Rightarrow$ Satisfies.

Now, assume 4 gaps

$\Rightarrow a + b + c + d = 11$, the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and $2 + 3 + 4 + 2 = 11$.

$\Rightarrow$ The possible cases for E are:

E	2010	2011	2012	2013	2014	2015
	1	3	5	3	1	-
	1	2	3	4	2	1
	1	2	4	3	2	1

In summary, the possible cases for all 5 companies is:

	2009	2010	2011	2012	2013	2014	2015	2016
A	1	2	3	4	5	3	2	1
	1	2	3	5	4	3	2	1
B	2012	2013	2014	2015				
	1	2	3	1				
	1	3	2	1				
C	2013	2014	2015	2016	2017			
	1	2	3	2	1			
D	2011	2012	2013	2014	2015			
	1	2	4	2	1			
E	2010	2011	2012	2013	2014	2015		
	1	3	5	3	1	-		
	1	2	3	4	2	1		
	1	2	4	3	2	1		

Given that total amount raised in 2014 is 12

$$\Rightarrow 3 + 3/2 + 2 + 2 + 1/2 = 12 \Rightarrow$$

$$\Rightarrow \text{possible case is } 3 + 3 + 2 + 2 + 2 = 12.$$

A) In 2013, B raised 2 crores and E also raised 3/4 crores $\Rightarrow$ Not Possible.

B) In 2013, A could have raised 5/4 and D raised 4 $\Rightarrow$ Possible.

C) In 2014, A raised 3 and B raised 3 $\Rightarrow$ Possible.

D) In 2014, B raised 3 where as E raised 2 $\Rightarrow 3 > 2 \Rightarrow$ Possible.

VIDEO SOLUTION

52. A

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Gowtham's Acquaintances are Ghinesh, Ezhil, Fani.

 VIDEO SOLUTION

53.3

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3 R4, R5 are $5 \times \text{means} = 17, 11, 19, 14$, and 17 respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18$, and 13 respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is 4 + 4)

=> We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ ($3 + 4$ or $4 + 3$) ----(1)

Consider R5, the missing elements should add up to $10 \Rightarrow 2 + 4 + 4$ or $4 + 3 + 3$ (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

Now, we can fill column R1 as $3 + 4$ and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

$\Rightarrow$ From the table, we can see that R1 gave a rating of 3 to Xavier.

[VIDEO SOLUTION](#)

54. C

x is the fraction of male members in the secondary section.

$$0.6 \cdot 80 + 0.55 \cdot 80 + 640x = 800 \cdot 0.475$$

$$\Rightarrow 0.6 + 0.55 + 8x = 4.75$$

$$\Rightarrow 8x = 3.6 \Rightarrow x = 0.45$$

$\Rightarrow$ 45% are males and 55% are females $\Rightarrow$ 288 males and 352 females

$\Rightarrow$ There are 144 vegetarian males and 208 vegetarian females .

Also there are 144 non-vegetarian males and 144 non-vegetarian females.

So leaving vegetarian females all others are same in nos.

[VIEW SOLUTION](#)

55. A

There are 640 secondary section students, 80 11th class students and 80 12th class students.

If the fraction of vegetarians in 12th class is x , then

$$80x + 80 \cdot 0.5 + 640 \cdot 0.55 = 0.53 \cdot 800$$

$$\Rightarrow x + 0.5 + 4.4 = 5.3 \Rightarrow x = 0.4$$

$\Rightarrow$ 40% are vegetarians in class 12.

[VIEW SOLUTION](#)

56. E

There are 640 secondary section students, 80 11th class students and 80 12th class students.

If the fraction of vegetarians in 12th class is x , then

$$80x + 80 \cdot 0.5 + 640 \cdot 0.55 = 0.53 \cdot 800$$

$$\Rightarrow x + 0.5 + 4.4 = 5.3 \Rightarrow x = 0.4$$

$\Rightarrow$ There are 32 vegetarian students in class 12.

25% of class 12 i.e 8 are males.

So there are 24 vegetarian female students.

No. of non-veg males = 40 and veg females = 24. Required answer $40 - 24 = 16$.

 [VIEW SOLUTION](#)

57. B

According to given condition No of male students in 11, 12 and in total are 44, 48 and 380.

So No of males in secondary are $380 - 44 - 48 = 288$.

288 students $\Rightarrow$ 45% of total secondary students.

 [VIEW SOLUTION](#)

58. 12

The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.

We have the reference point of P1 at 66

Trying to fill in for C1:

The possible values are 75, 80, 85, 90, 95

Since 12 grams is of other nutrients, we can eliminate 90 and 95.

If we take 85, $85 + 12 = 97$, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, $75 + 12 = 87$ would leave 13 grams of protein, which, too, is eliminated.

Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.

Taking 85 carbs would give 2 grams of protein, which can be eliminated.

80 and 70 would also not work for the same reason.

Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100 - 65 - 14 - 8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100 - 70 - 14 - 8 = 8$ grams of protein.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100 - 7 - 16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Carbohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

We can see that there were 12 grams of protein in M2.

[VIDEO SOLUTION](#)

59. D

From given conditions we find that :

	London(l)	Paris(p)	Beijing(b)	NY(n)	Total
Round1(r1)	30	24	16	12	82
Round2(r2)	30	24(p r1) +3 (n r1) +4 (B r1) +1(n - representative)	12 (b r1) +9(n r1)	0	83
Round3(r3)	30+8 (Beijing r1 , r2)	24(p r1,r2) +4(b r1 ;p r2) +4(b r1,r2) +3 (n r1, p r2) +1 (n-representative) +1(b-representative)	0	0	75

Required value is $8 \times 100 / 12 = 66.67\%$. Hence option D.

[VIDEO SOLUTION](#)

60. C

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be $500000 - 275000 - 95000 - 85000 = 45000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 $\Rightarrow$ the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

From 3, Let the total votes in D be $100x \Rightarrow$ The winning candidate in constituency D must have got $37500 + 5x$.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	$100x$
No. of votes polled by the winning candidate	2,75,000	48,750	140006	$37500 + 5x$
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

The candidates who didn't lose the deposit must have got $<16.67\%$ $\Rightarrow$ 3rd runner up must surely didn't get the deposit.

Also, the candidates who got security deposit must have got 65% of votes.

Case I:

Let top three candidates got the security deposit $\Rightarrow 37500+5x+37500+30000 = 65x \Rightarrow x=1750 \Rightarrow 100x=175000$

Case II:

Let top three candidates got the security deposit $\Rightarrow 37500+5x+37500 = 65x \Rightarrow 60x = 1250 \Rightarrow x=125000$ but 16.66% of $125000 = 20834 \Rightarrow$ 2nd runner up must got security deposit. So, this case is not valid.

For all the constituencies let's look at the candidates who lost their security deposit.

A $(500000-275000-95000-85000)=45000$.

B $(325000-48750)=276250$

C (0) and D $(61250)=175000-46250-30000-37500$

Hence percentage will be $382500/1600000 \times 100 = 23.91\%$

 VIDEO SOLUTION

CAT Percentile Predictor

61. C

It's given in the question that the first runner up polled 10,000 more votes than the second runner up in constituency A. Now the first runner up has got 95000 votes, hence the second runner up will get 85000 votes.

Now the remaining votes will be $500000-275000-95000-85000=45000$

From 2, None of the candidates who contested in constituency C lost their security deposit. The difference in votes polled by any pair of candidates in this constituency was at least 10,000 $\Rightarrow$ the person who got 5th highest votes must have got $> 600030/6 \Rightarrow \geq 100006$. Since it is also given that the difference of votes is ≥ 10000 , the only possible case is winner, 1st runner up, 2nd runner up, 3rd runner up, 4th runner up must have got 140006, 130006, 120006, 110006, 100006 respectively which sums upto exactly 600030.

From 3, Let the total votes in D be $100x \Rightarrow$ The winning candidate in constituency D must have got $37500+5x$.

The table now looks like:

	Constituency			
	A	B	C	D
No. of candidates contesting	10	12	5	8
Total No. of valid votes polled	5,00,000	3,25,000	6,00,030	$100x$
No. of votes polled by the winning candidate	2,75,000	48,750	140006	$37500+5x$
No. of votes polled by the first runner up	95,000		130006	37,500
No. of votes polled by the second runner up	85000		120006	30,000
% of valid votes polled by the third runner up			16.67%	10%

The candidates who didn't lose the deposit must have got $<16.67\%$ $\Rightarrow$ 3rd runner up must surely didn't get the deposit.

Also, the candidates who got security deposit must have got 65% of votes.

Case I:

Let top three candidates got the security deposit $\Rightarrow 37500+5x+37500+30000 = 65x \Rightarrow x=1750 \Rightarrow 100x=175000$

Case II:

Let top three candidates got the security deposit $\Rightarrow 37500+5x+37500 = 65x \Rightarrow 60x = 1250 \Rightarrow x=125000$ but 16.66% of $125000 = 20834 \Rightarrow$ 2nd runner up must got security deposit. So, this case is not valid.

The increasing order C will always come after D which is not happening in the third option. Hence that is the correct answer.

VIDEO SOLUTION

62. B

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a**.

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b**.

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectively.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60**.

From 5, $s+c+t+c+70=306 \Rightarrow 2c=96 \Rightarrow c=48$.

From 4, $(b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 \Rightarrow 54+m+n-2a-b=40 \Rightarrow 2a+b=14+m+n \Rightarrow 2b=28+2m+2n-4a...$
(i)

From 7, $b-a=n-b \Rightarrow 2b=a+n...$ (ii) and $m-b=54-a \Rightarrow a+m=54+b \Rightarrow 2b=2a+2m-108...$ (iii)

From 8, $108=b+n \Rightarrow 2b=216-2n...$ (iv)

from (i) & (iii), $28+2m+2n-4a=2a+2m-108 \Rightarrow 6a=136+2n \Rightarrow 3a=68+n...$ (v)

from (ii) & (iv), $216-2n=a+n \Rightarrow a+3n=216 \Rightarrow 3a=648-9n...$ (vi)

from (v) & (vi), $648-9n=68+n \Rightarrow n=58 \Rightarrow a=42 \Rightarrow b=50 \Rightarrow m=62$.

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

∴ total sales amount, in Crore Rupees, in 2019 for the chain of departmental stores =900 cr.

63. D

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a**.

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b**.

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectively.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60**.

From 5, $s+c+t+c+70=306 \Rightarrow 2c=96 \Rightarrow c=48$.

From 4, $(b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 \Rightarrow 54+m+n-2a-b=40 \Rightarrow 2a+b=14+m+n \Rightarrow 2b=28+2m+2n-4a...$
(i)

From 7, $b-a=n-b \Rightarrow 2b=a+n...$ (ii) and $m-b=54-a \Rightarrow a+m=54+b \Rightarrow 2b=2a+2m-108...$ (iii)

From 8, $108=b+n \Rightarrow 2b=216-2n...$ (iv)

from (i) & (iii), $28+2m+2n-4a=2a+2m-108 \Rightarrow 6a=136+2n \Rightarrow 3a=68+n...$ (v)

from (ii) & (iv), $216-2n=a+n \Rightarrow a+3n=216 \Rightarrow 3a=648-9n...$ (vi)

from (v) & (vi), $648-9n=68+n \Rightarrow n=58 \Rightarrow a=42 \Rightarrow b=50 \Rightarrow m=62$.

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

Hence the percentage change is 50.

64. A

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a**.

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b**.

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectively.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60**.

From 5, $s+c+t+c+70=306 \Rightarrow 2c=96 \Rightarrow \mathbf{c=48}$.

From 4, $(b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 \Rightarrow 54+m+n-2a-b=40 \Rightarrow 2a+b=14+m+n \Rightarrow 2b=28+2m+2n-4a...$
(i)

From 7, $b-a=n-b \Rightarrow 2b=a+n...$ (ii) and $m-b=54-a \Rightarrow a+m=54+b \Rightarrow 2b=2a+2m-108...$ (iii)

From 8, $108=b+n \Rightarrow 2b=216-2n...$ (iv)

from (i) & (iii), $28+2m+2n-4a=2a+2m-108 \Rightarrow 6a=136+2n \Rightarrow 3a=68+n...$ (v)

from (ii) & (iv), $216-2n=a+n \Rightarrow a+3n=216 \Rightarrow 3a=648-9n...$ (vi)

from (v) & (vi), $648-9n=68+n \Rightarrow \mathbf{n=58} \Rightarrow \mathbf{a=42} \Rightarrow \mathbf{b=50} \Rightarrow \mathbf{m=62}$.

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

Increase in sales =12

VIDEO SOLUTION

65. D

From 1, Let sales amounts in the Apparel departments for Delhi and Kolkata in 2018 be **a**.

From 2, Let sales amounts in the Apparel departments for Mumbai and Bengaluru in 2018 and for Delhi be **b**.

From 3, let the sales amounts in the Home Décor departments for Mumbai and Kolkata in 2018 be **c** and that of Delhi and Bengaluru be **s** and **t** respectively.

Let the sales amounts in the Apparel departments for Mumbai and Bengaluru in 2019 be **m** and **n** respectively.

The table looks like:

Sales Amounts (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparel	a	b	b	m	b	n	a	54
Electronics	78	98	82	102	90	70	80	100
Home Décor	s	100	c	72	t	80	c	54

From 6, **s=80 and t=60**.

From 5, $s+c+t+c+70=306 \Rightarrow 2c=96 \Rightarrow c=48$.

From 4, $(b-a)+(m-b)+(n-b)+(54-a)=(20)+20-20+20 \Rightarrow 54+m+n-2a-b=40 \Rightarrow 2a+b=14+m+n \Rightarrow 2b=28+2m+2n-4a$...

(i)

From 7, $b-a=n-b \Rightarrow 2b=a+n$...(ii) and $m-b=54-a \Rightarrow a+m=54+b \Rightarrow 2b=2a+2m-108$...(iii)

From 8, $108=b+n \Rightarrow 2b=216-2n$...(iv)

from (i) & (iii), $28+2m+2n-4a=2a+2m-108 \Rightarrow 6a=136+2n \Rightarrow 3a=68+n$...(v)

from (ii) & (iv), $216-2n=a+n \Rightarrow a+3n=216 \Rightarrow 3a=648-9n$...(vi)

from (v) & (vi), $648-9n=68+n \Rightarrow n=58 \Rightarrow a=42 \Rightarrow b=50 \Rightarrow m=62$.

The final table looks like this

Sales Amount (Crore Rupees)								
	Delhi		Mumbai		Bengaluru		Kolkata	
	2018	2019	2018	2019	2018	2019	2018	2019
Apparels	42	50	50	62	50	58	42	54
Electronics	78	98	82	102	90	70	80	100
Home Decor	80	100	48	72	60	80	48	54

VIDEO SOLUTION

[Important Verbal Ability Questions for CAT \(Download PDF\)](#)

66.24

The set's starting point is that the sum of each row must be 100

Clue 3 tells us that all the missing elements in the Carbs column are multiple of 5. Similarly, clue 4 tells us that the missing elements of the other three columns are multiples of 4.

Note that we look at clue 1, which says that the carbs in C1 and C2 must be more than any carbs in any pseudo cereal.

We have the reference point of P1 at 66

Trying to fill in for C1:

The possible values are 75, 80, 85, 90, 95

Since 12 grams is of other nutrients, we can eliminate 90 and 95.

If we take 85, $85+12 = 97$, which would leave 3 grams of protein, but from clue 4, we know this has to be a multiple of 4.

Taking 75, $75+12 = 87$ would leave 13 grams of protein, which, too, is eliminated.

Leaving only 80 grams of carb in C1 and 8 grams of protein.

Similar logic is to be applied for C2; we have 13 grams already present, so the values of 90 and 95 are eliminated.

Taking 85 carbs would give 2 grams of protein, which can be eliminated.

80 and 70 would also not work for the same reason.

Leavin has only 75 grams of carbs, leaving 12 grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2		14		8

Using clues 1 and 2 together, we can determine that the carbs in P2 must be less than 75 but at least greater than 62.

The only possible values are 65 and 70

Putting 65 grams of carbs in P2 gives us $100-65-14-8 = 13$ grams of protein, which is invalid.

Putting 70 grams of carbs in P2 gives $100-70-14-8 = 8$ grams of protein.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2			7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

Clue 1 gives us that the protein in M2 should be less than that present in either of the pseudo-cereals, which we right now have a baseline of 14.

So the protein in M2 (and M3) can be 12, 8, 4, 0

The protein and carbs in M2 should add up to $100-7-16 = 77$

This is only possible if the protein count is 12, giving carbs as 65

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56		12	
Pseudo-cereals	P1	66			10
	P2	70	14	8	8

The protein in M3 can be 0, 4, 8 or 12

We are given in clue five that the protein in P1 is double that in M3.

The protein in P1 thus can be 0, 8, 16 or 24

Since this P1 protein also has to be more than M1 and M2 protein, it can not be 0 or 8

Leaving only 16 or 24 as the valid values.

The protein and fats in P1 must add up to $100 - 66 - 10 = 24$

If P1 had 24 grams of protein, then it would have 0 grams of fat, but in clue 4, we are given that all missing fats are **non-zero multiples of 4**

Hence, the only possible protein value in P1 is 16, with 8 grams of fats. Giving 8 grams of protein in M3 and 24 grams of others in M3.

		Cabohydrates	Protein	Fats	Others
Cereal	C1	80	8	0	12
	C2	75	12	3	10
Millets	M1	62	10		
	M2	65	12	7	16
	M3	56	8	12	24
Pseudo-cereals	P1	66	16	8	10
	P2	70	14	8	8

There were 24 grams of other nutrients in M3.

[VIDEO SOLUTION](#)

67. A

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$$\Rightarrow 60x = 24$$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*0.4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+ 40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $\{66-70*(0.6)\}/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the weight of the test component is 0.6

The correct option is A

 VIDEO SOLUTION

68. A

Since A, B, C are mutual strangers, (B, C) are strangers for A, (A, C) are strangers for B, (A, B) are strangers for C.

Since the total number of strangers+ acquaintances+ friends for any among the 7 is 6.

The number of friends for Amudha is 1, the number of friends for Gowtham is 1.

Using statement 3 Chandran and Gowtham are friends.

Using statement 2: Amudha, Dinesh, and Fani are Ezil's friends. Similarly, Ezil is a friend of Amudha, Dhinesh, and Fani.

Using statement 4 Every friend of Bharatan is an acquaintance of Amudha, and every acquaintance of Amudha is a friend of Bharatan, Hence the number of acquaintances of Bharatan is equal to the number of friends of Amudha.

Using statement 5 Every friend of Amudha is an acquaintance of Bharatan, and every acquaintance Bharatan is a friend of Amudha, Hence the number of acquaintances of Amudha is equal to the number of friends of Bharatan.

Hence Bharatan has one friend, 1 Acquaintance, 4 strangers.

For Amudha we are yet to find a relationship with Dhinesh, Fani, and Gowtham. Any among the three can be the stranger for Amudha, considering the three different cases.

Case 1 :

Considering Fani as an acquaintance of Amudha, then Dhinesh and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Fani is an acquaintance of Amudha, hence Amudha is an Acquaintance of Fani, Fani is a friend of Bharathan and hence Bharathan is a friend of Fani. But Fani has only one friend and Ezhil is already a friend of Fani.

Hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1, (F)	4 (B, C, D, G)
Bharatan	1, (F)	1, (E)	4, (A, C, D, G)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 2 :

Considering Gowtham as an acquaintance of Amudha, then Dhinesh and Fani are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan. Since Gowtham is a friend of Bharatan, Bharatan must be a friend of Gowtham. But Gowtham can only have one friend and it already mentioned that Chandran is a friend of Gowtham and hence this case fails.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (G)	4 (B, C, D, F)
Bharatan	1, (G)	1, (E)	4, (A, C, D, F)
Chandran	G	1	(A, B)
Dhinesh	E		2
Ezhil	A, D, F		1
Fani	1 (E)		
Gowtham	1 (C)	3	2

Case 3 :

Considering Dhinesh as an acquaintance of Amudha, then Fani and Gowtham are strangers to Amudha.

Every acquaintance of Amudha is a friend of Bharathan and since the friend and acquaintance of Bharatan are known the strangers are found for Bharatan.

Since Fani, Gowtham are strangers to Amudha, Bharatan. Amudha, Bharatan are strangers to Fani, Gowtham.

The 2 strangers to Gowtham and his only friend are known. Hence his three acquaintances are (Dhinesh, Ezhil, Fani).

Hence Gowtham is an acquaintance of Dhinesh, Ezhil, and Fani.

Dhinesh is an acquaintance of Amudha and hence Amudha must be an acquaintance of Dhinesh.

Dhinesh is a friend of Bharatan and hence Bharatan is a friend of Dhinesh.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4 (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	G	1	(A, B)
Dhinesh	E, B	G, A	2
Ezhil	A, D, F	G	1
Fani	1 (E)	G	A, B
Gowtham	1 (C)	3 (D, E, F)	2 (A, B)

Ezhil is an acquaintance of Bharatan and hence Bharatan is an acquaintance of Ezhil. The only stranger to Ezhil who is left is Chandran. Hence Ezhil is a stranger to Chandran.

The two strangers to Dhinesh who are left are Chandran and Fani. Chandran is a stranger to Dhinesh and hence Dhinesh is a stranger to Chandran.

The only acquaintance of Chandran who is left is Fani.

The remaining relationships with Fani are Dhinesh and Chandran. Dhinesh is a stranger to Fani and Chandran is an acquaintance of Fani.

	No of friends	No of Acquaintances	No of Strangers
Amudha	1, (E)	1 (D)	4, (B, C, F, G))
Bharatan	1, (D)	1, (E)	4, (A, C, F, G)
Chandran	1, (G)	1 (F)	4, (A, B, E, D)
Dhinesh	2, (E, B)	2, (G, A)	2, (C, F)
Ezhil	3, (A, D, F)	2, (G, B)	1, (C).
Fani	1 (E)	2, (G, C)	3, (A, B, D)
Gowtham	1 (C)	3 (D, E, F)	2, (A, B)

Dhinesh is an acquaintance of Amudha

[VIDEO SOLUTION](#)

69. D

Since tara received the same grade in exactly 3 subjects , only one possibility is getting B in 3 subjects and F in other 2. So in operation her grade would be either B or F.Among options only manab has B in operations . hence option D.

[VIDEO SOLUTION](#)

70. 120

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => x = 40 (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The total number of registrations in April is 120.

 VIDEO SOLUTION

Data Interpretation for CAT Questions (download pdf)

71.A

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => $x = 40$ (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

Total registrations in Jan = Apr = 120 and Feb = May = 130.

 VIDEO SOLUTION

72.40

Given that in every month, both online and offline registration numbers were multiples of 10.

From (2), in Jan, the number of offline registrations was twice that of online registrations.

=> If x is number of online registrations => 2x is the number of offline registrations => 3x is the total number of registrations.

According to the data given in the table => 3x should lie between the minimum and maximum total number of registrations. => $x = 40$ (as x should also be a multiple of 10)

=> In Jan => (40,80) are the online and offline registrations respectively.

Similarly from (3) => In Apr (80,40) are the online and offline registrations respectively.

From-5, the number of online registrations is highest in may => In may there are 100 online registrations. The lowest possible number of offline registrations is 30 and maximum possible total registrations is 130 => In May (100,30) are the online and offline registrations respectively.

Let us assume, 'x' to be the number of offline registrations in May = number of online registrations in March.

Let us capture all this data in a table:

Month	Online	Offline	Total
Jan	40	80	120
Feb	y	x	
Mar	x	z	
Apr	80	40	120
May	100	30	130

From the table given in the question, 50 is the median for Offline data

=> x should lie between 50 and 80 (included)

For 80 to be the median for the online data => y lie between 80 and 100 (included).

Now, consider Feb => Minimum value of $y + x = 80 + 50 = 130$ (which is the maximum value possible of the total possible registrations)

=> $x = 50$ and $y = 80$

Since, 110 is the minimum number of total registrations, the only possibility is in March => $50 + z = 110$ => $z = 60$.

Now, filling the complete table we get,

Month	Online	Offline	Total
Jan	40	80	120
Feb	80	50	130
Mar	50	60	110
Apr	80	40	120
May	100	30	130

The number of online registrations in Jan is 40.

 VIDEO SOLUTION

73. A

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$$\Rightarrow 60x = 24$$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*0.4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $\{66-70*(0.6)\}/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the maximum aggregate score obtained is 68. The correct option is A

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74.40

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$$\Rightarrow 60x = 24$$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*0.4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $\{66-70*(0.6)\}/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score ($T*0.6+P*0.4$)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

Hence, the score obtained by Mathew in the test is 40

 VIDEO SOLUTION

75. C

It is given that there are only three female students - Amala, Koli, and Rini - and only three male students - Biman, Mathew, and Shyamal - in a course.

It is also known that the aggregate score in the course is a weighted average of the two components, with the weights being positive and adding to 1.

Let the project score component be x , which implies the test score component will be $(1-x)$. The projects are done in groups of two, with each group consisting of a female and a male student, which implies there are three groups for the project. It is also known that both the group members obtain the same score in the project. The score obtained in the project is 40, 60, and 80, respectively.

Therefore, we can say that each female student will consist of a different group, and no two male students or female students will be in the same group.

For the test scores, there are six scores given for six students among which four are distinct and the remaining two are average scores, which is 60. It is also known that the maximum score possible is 80, and the minimum score is 40.

Hence, the distinct scores are 80, 70, 50, and 40 (since all the test scores are multiple of 10), and the remaining two scores are 60, and 60, respectively.

From point 3, we know that Amala's score in the project was double that of Koli in the same, but Koli scored 20 more than Amala in the test. Hence, we can say the score obtained by Amala in the project is 80, and the score obtained by Koli is 40, which implies the score obtained by Rini in the project is 60. Now, Koli scored 20 more than Amala in the test, which implies the score obtained by Koli can be either 80, 70, or 60.

The score obtained by them is given below:

Students	Test scores	Project scores
Amala	40/50/60	80
Koli	60/70/80	40
Rini		60
Biman		
Mathew		
Shyamal		

It is known that Amala had the highest aggregate score, and Shyamal scored the second highest on the test. He scored two more than Koli, but two less than Amala in the aggregate.

Hence, the score obtained by Shyamal in the test is 70, which implies Koli can't score 70 in the test => Amala can't score 50 in the test.

Students	Test scores	Project scores
Amala	40/60	80
Koli	60/80	40
Rini		60
Biman		
Mathew		
Shyamal	70	

It is given that Shyamal scored two more than Koli, but two less than Amala in the aggregate. Hence, the aggregate score of Amala is 4 more than Koli. It is also known that Amala had the highest aggregate score.

Case 1: The test score of Amala is 40

Students	Test scores	Project scores	Aggregate score
Amala	40	80	$40(1-x)+80x$
Koli	60	40	$60(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $40(1-x)+80x = 60(1-x)+40x+4$

$$\Rightarrow 60x = 24$$

$$\Rightarrow x = 0.4$$

Hence, the aggregate score obtained by Amala is $40(1-0.4)+80*0.4 = 56$

The minimum aggregate score of Shyamal is $70(1-0.4)+ 40*0.4 = 58$, which is greater than Amala.

Hence, **Case 1 is not possible.**

Hence, the table is given below:

Students	Test scores	Project scores	Aggregate score
Amala	60	80	$60(1-x)+80x$
Koli	80	40	$80(1-x)+40x$
Rini		60	
Biman			
Mathew			
Shyamal	70		

Therefore, $60(1-x)+80x = 80(1-x)+40x+4$

$$\Rightarrow 60+20x = 84-40x$$

$$\Rightarrow 60x = 24 \Rightarrow x = 0.4$$

Hence, the aggregate score of Amala is $60(1-0.4)+80*0.4 = 68$, which implies the aggregate score of Shyamal is $(68-2) = 66$

Hence, the score obtained by Shyamal in Project is $\{66-70*(0.6)\}/0.4 = 60$.

It is also known that Biman scored second lowest in the test, which implies the score of Biman in the test is 50, and he scored the lowest in the aggregate. It is also known that Mathew scored more than Rini in the project, but less than her in the test. Hence, Mathew scored 80 in the project (since Rini scored 60 in the project), and Biman scored 40 in the project.

Similarly, Rini Scored more than Mathew on the test, which implies the score obtained by Rini is 60, and the score obtained by Mathew is 40 in the test.

Hence, the final table will look like this:

Students	Test scores (T)	Project scores(P)	Aggregate score (T*0.6+P*0.4)	Project Pair
Amala	60	80	68	Amala, Mathew
Koli	80	40	64	Koli, Biman
Rini	60	60	60	Rini, Shyamal
Biman	50	40	46	Biman, Koli
Mathew	40	80	56	Mathew, Amala
Shyamal	70	60	66	Shyamal, Rini

From the table, we can see that (Amala, Mathew), (Koli, Biman), and (Shyama, Rini) are the three groups for the project.

Hence, the correct option is C

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